

SHUBHAGATA BHAUMIK

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EXPERIENCE

Post Doctoral Fellow

Indian Institute of Technology (IIT) Bombay

Jan. 2026 - Current

Mumbai, India

EDUCATION

PhD, University of Florida, *Gainesville, FL, USA*

2019 - 2025

Search for Eccentric Binary Black Hole Mergers with Gravitational Waves

Major: Physics; Supervisor: Prof. Imre Bartos

GPA: 3.97/4.00

BS-MS Dual Degree, Indian Institute of Science Education and Research, *Kolkata, India*

2014 - 2019

Major: Physics; Supervisor: Prof. Rajesh Kumble Nayak

GPA: 8.5/10.0

RESEARCH INTERESTS

Gravitational wave data analysis, astrophysical interpretations of black hole populations, machine learning applications to gravitational wave physics, multi-messenger astrophysics.

RESEARCH EXPERIENCE

Graduate Assistant

2020 - 2025

University of Florida (UF)

- Member of a large scientific collaboration (LIGO-Virgo-KAGRA) and leading the collaboration's efforts on the search for gravitational waves from binary black holes with eccentric orbits.
- Developed a search for eccentric binary black hole mergers using modeled and unmodeled algorithms.
- Source property extraction of gravitational wave signals from binary black holes with eccentric orbits using Markov-chain Monte Carlo methods.
- Worked on a computationally effective machine-learning-based approach to distinguish between binaries formed through isolated and dynamic formation scenarios.

Master's Thesis

May 2018 - May 2019

Indian Institute of Science Education and Research, Kolkata

- Carried out research on the applications of particle swarm optimization algorithm in gravitational wave data analysis. In particular, I worked on improving the performance of the algorithm for detection of weak signals.
- Incorporated machine learning techniques in the implementation of Particle Swarm Optimization.

HONORS & AWARDS

Charles Vincent and Heidi Cole McLaughlin Dissertation Fellowship

2025

UF College of Liberal Arts and Sciences Travel Grant

2024

Association for Academic Women's (AAW) Emerging Scholar Award

2024

American Astronomical Society (AAS) FAMOUS Travel Grant

2024

Charles F. Hooper Jr. Memorial Award

2023

Awarded to senior graduate students in physics who have shown distinction in research and teaching.

UF College of Liberal Arts and Sciences Travel Grant

2023

Steigleman Fellowship

2022

Awarded to a graduate student for notable achievements in the field of astrophysics.

Institute for High Energy Physics and Astrophysics Fellowship

2020

Awarded to a first year graduate student pursuing research in high energy physics or astrophysics, selected by a committee of faculty in Institute for High Energy Physics and Astrophysics.

INSPIRE Fellowship

2014 - 2019

Awarded to students who are among the top 1% of all students appearing for 12th grade examinations and are pursuing courses in Natural Sciences during BS or MS, awarded by Department of Science and Technology, Govt. of India.

PRESENTATIONS

Invited Presentations

- University of Wisconsin Milwaukee, *Virtual* Apr. 2025
Search for Eccentric Binary Black Hole Mergers with Gravitational Waves
- University of Warsaw, *Warsaw, Poland* Oct. 2024
Expanding the Gravitational Wave Frontier: Detecting Exceptional Binary Black Hole Mergers with a Model-Independent Search
- University of Texas-Austin, *Austin, TX, USA* June 2024
Search for Eccentric Black Holes Coalescences with Gravitational Waves
- University of Miami Physics Conference, *Miami, FL, USA* Dec. 2023
Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo
- University of Miami Physics Conference, *Miami, FL, USA* Dec. 2022
Search for Eccentric Binary Black Hole Mergers with Gravitational Waves

Contributed Presentations

- LIGO-Virgo-KAGRA Collaboration September Meeting, *Barcelona* Sept. 2024
- LIGO-Virgo-KAGRA Collaboration March Meeting, *Baton Rouge, LA* March 2024
- Graduate Student and Postdoc Seminar, *UF Physics* Feb. 2024
- American Astronomical Society (AAS) 243rd Winter Meeting, *New Orleans, LA* Jan. 2024
- LIGO-Virgo-KAGRA Collaboration September Meeting, *Virtual* Sep. 2023
- American Physical Society April Meeting, *Minneapolis, MN* April 2023
- LIGO-Virgo-KAGRA Collaboration March Meeting, *Evanston, IL* March 2023
- LIGO-Virgo-KAGRA Collaboration September Meeting, *Virtual* Sep. 2022
- Graduate Student and Postdoc Seminar, *UF Physics* Oct. 2021
- LIGO-Virgo-KAGRA Collaboration September Meeting, *Virtual* Sep. 2021

COLLABORATION PAPERS

Collaboration-wide publication where I made major contributions.

LIGO-Virgo Collaboration [including **S. Bhaumik** as **lead analyst**], *Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo*, ApJ. 973 (2024) 2, 132, arXiv:2308.03822

SHORT AUTHOR-LIST PUBLICATIONS

T. RoyChowdhury, V. Gayathri, R. Gamba, **S. Bhaumik**, I. Bartos, J. Creighton, *The Cost of Circularity: Quantifying Eccentricity-Induced Biases in Binary Black Hole Inference*, submitted to PRD, arXiv:2603.02453 (2026).

V. Gayathri, Giuliano Iorio, ... **S. Bhaumik**,... et al., *Reconstructing the origin of black hole mergers using sparse astrophysical models*, submitted to A&A, arXiv:2509.09647 (2025).

S. Bhaumik, V. Gayathri, I. Bartos, J. Anglin, G. Carullo, J. Healy, S. Klimenko, J. Lange, C. Lousto, T. Mishra, M. J. Szczepańczyk, *Gravitational Wave Detector Sensitivity to Eccentric Black Hole Mergers*, Phys. Rev. D 111 12 123032 (2025), arXiv:2410.15192.

T. Mishra, **S. Bhaumik**, V. Gayathri, M. J. Szczepańczyk, I. Bartos, S. Klimenko, *Gravitational Waves Detected by a Burst Search in LIGO/Virgo’s Third Observing Run*, Phys. Rev. D 111 2 023054 (2025), arXiv:2410.15191.

M. J. Szczepańczyk, Yanyan Zheng,... **S. Bhaumik**,... et al. *Optically targeted search for gravitational waves emitted by core-collapse supernovae during the third observing run of Advanced LIGO and Advanced Virgo*, Phys. Rev. D 110 4, 042007 (2024), arXiv:2305.16146.

H. L. Iglesias, J. Lange, I. Bartos, **S. Bhaumik**, R. Gamba, V. Gayathri, A. Jan, R. Nowicki, R. O’Shaughnessy, D. Shoemaker, R. Venkataramanan, K. Wagner, *Eccentricity estimation for five binary black hole mergers with higher-order gravitational wave modes*, ApJ. 972 (2024) 1, 65, arXiv:2208.01766.

M. Szczepańczyk, F. Salemi, S. Bini, T. Mishra, G. Vedovato, V. Gayathri, I. Bartos, **S. Bhaumik**, M. Drago, O. Halim, C. Lazzaro, A. Miani, E. Milotti, G. Prodi, S. Tiwari, S. Klimenko, *Search for gravitational-wave bursts in the third Advanced LIGO-Virgo run with coherent WaveBurst enhanced by machine learning*, Phys. Rev. D 107 6, 062002 (2023).

T. Mishra, B. O’Brien, M. Szczepańczyk, G. Vedovato, **S. Bhaumik**, V. Gayathri, G. Prodi, F. Salemi, E. Milotti, I. Bartos, S. Klimenko, *Search for binary black hole mergers in the third observing run of Advanced LIGO-Virgo using coherent WaveBurst enhanced with machine learning*, Phys. Rev. D 105, 083018 (2022).

T. Mishra, B. O’Brien, V. Gayathri, M. Szczepańczyk, **S. Bhaumik**, I. Bartos, S. Klimenko, *Optimization of model independent gravitational wave search for binary black hole mergers using machine learning*, Phys. Rev. D 104, 023014 (2021).

WHITE PAPER CONTRIBUTIONS

I. Bartos, **S. Bhaumik**, et al., *Orbital Eccentricity as the Fingerprint of Merger Origin* [link]

ACADEMIC SERVICES

<i>Service within LIGO-Virgo-KAGRA Collaboration</i>	
Member, Eccentric Parameter-Estimation Task Force	Oct. 2025 - Present
Reviewer, TEOBResumS-DALI (eccentric) waveform	Dec. 2023 - May. 2025
Rota member, Online Parameter Estimation for Gravitational-wave Candidates in LIGO-Virgo-KAGRA’s Fourth Observing Run	Aug. 2023 - Dec. 2024
Science Summary Author, Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo [link]	2023
<i>At University of Florida</i>	
Organizer, Graduate Student and Postdoc Seminars in Astrophysics	Aug. 2021 - Dec. 2022

OUTREACH & MENTORING

One-on-one Mentoring for Graduate Students, Physics Graduate Community, UF	Sep. 2022 - Aug. 2025
Guest Lecturer, Lincoln Middle School, Gainesville	Oct. 2022
Member, UF Physics Inclusivity, Equity and Diversity Alliance (IDEA)	Aug. 2021 - Dec. 2023
Research Mentor	
- Urja Shah [Undergraduate student → Georgia Tech]	Aug. 2020 - May 2021
- Jeremiah Anglin [Graduate Student]	Aug. 2023 - Aug. 2025
- Hudson Shields [Undergraduate Student]	Aug. 2025 - Present

TEACHING EXPERIENCE

Graduate-level Electromagnetism [Grader]	Fall 2024
Graduate-level Classical Mechanics [Grader]	Fall 2023, Spring 2024, Fall 2024
Graduate-level Quantum Mechanics [Grader]	Fall 2023, Spring 2024, Fall 2024
Mechanics 1	Spring 2024

Physics 1 [Online]

Fall 2023

Physics 1

Fall 2022, Spring 2023

Introductory Physics Laboratory [Online]

Fall 2020, Spring 2021

Introductory Physics Laboratory

Fall 2019, Spring 2020

REFERENCES

Dr. Imre Bartos

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